ESTIMATION :-

→ Estimation is a process of drawling conclusion for population based on sample data.

→ ESTIMATION TYPES :-

- → POINT ESTIMATION
- → INTERVAL ESTIMATION

SAMPLE STATISTICS :-

→ A sample statistic is a numerical numbers calculated from sample data (e.g., sample mean, sample proportion, sample standard deviation).

→ It changes from sample to sample and is used to estimate the unknown population parameter.

POPULATION PARAMETER :-

→ A population parameter is a numerical value that describes a characteristic of an entire population (for example, population mean 'μ', population proportion 'P', population standard deviation σ).

→ It is usually fixed but unknown in practice, so it is estimated using sample statistics).

POPULATION PARAMETER	SAMPLE STATISTICS
POPULATION MEAN - μ	SAMPLE MEAN - $\bar{x}$
POPULATION S.D. - σ	SAMPLE S.D. - s
POPULATION PROPORTION - P	SAMPLE PROPORTION - p
POPULATION SIZE - N	SAMPLE SIZE - n
POPULATION DATA VALUE - X	SAMPLE DATA VALUE - x
CORRELATION COEFFICIENT - ρ	CORRELATION COEFFICIENT - r

Point Estimation

- It is the simplest form of estimation
- It is the process of giving the single best value as your guess for the unknown population parameter.

→ Properties of Point Estimation :-

1. Consistent

- When your sample size is small, your estimate may be far from the true values.
- As you take more and more data, the estimate moves closer to the true value.

In short :-

- More data ⟶ Better Estimate

Eventually, estimate ≈ True Value

Example : Actual/True mean = 28

- Sample Size = 10 → μ = 25
- Sample Size = 100 → μ = 27
- Sample Size = 1000 → μ = 27.8

∴ The estimator is consistent.

2.

EFFICIENT [illegible] ESTIMATOR

2. UNBIASED AND [illegible]

- An estimator is unbiased if it does not consistently favour one side.

- Sometimes estimate is above the true value
- Sometimes below
- But on average, it hits the true value.

- Think of the arrow in the picture:

  28 ⟷ 25

  27.5 ⟷ 28

→ Estimates are scattered around the true value, not shifted towards one side.

* KEY DIFFERENCE

- Consistent cares about sample size.
  "With more data, I improve"

- Unbiased cares about average correctness.
  "I don't cheat in one direction"

METHODS TO FIND POINT ESTIMATION

1. MLE (MAXIMUM LIKELIHOOD ESTIMATION)
2. LAPLACE ESTIMATION
3. WILSON ESTIMATION
4. JEFFREY ESTIMATION

* FORMULAE :- In all formulae we need 3 values :-
  - S → NUMBER OF SUCCESS
  - T → NUMBER OF TRIALS
  - Z → Z-SCORE

1. MLE :- S/T

2. LAPLACE :- $(S+1)/(T+2)$

3. WILSON :- $(S+Z^2/2)/(T+Z^2)$

4. ZEFFREY :- $(S+0.5)/(T+1)$

DRAWBACKS OF POINT ESTIMATE :-

- PROBLEM OF RELIABILITY
  NOT ENOUGH EVIDENCE TO SUPPORT THAT THE POINT ESTIMATE would lie near to the POPULATION PARAMETER

→ THEREFORE TO counter this drawback of point estimation we use INTERVAL ESTIMATION.

Interval Estimation

- Provides intervals in which population parameter falls.
- More accurate than Point Estimation as it provides Range.